Pre-study Protocol: Selection and Validation of Emotional Video Stimuli

Aim of the Study

Our aim was to create a database of audio-visual content designed to induce different types of emotions. For this purpose, we selected a wide range of videos from commercial movies and series, and social media content, to be assessed according to several features: valence, arousal, label, and intensity of the induced emotion.

Twenty independent participants rated a pre-selected sample of 82 videos. As a result, we obtained a sample of 43 videos likely to induce nine kinds of emotions.

Method

Pre-selection

Six graduates in Psychology (1 engineer and 5 bachelor's students), all female, under 30 years old, and French native speakers, were asked to provide audio-visual content thought likely to induce emotion, based on their knowledge and personal sensibility. We suggested a list of emotional categories to guide the search (admiration, amusement, anger, anxiety, contempt, despair, disgust, fear, happiness, interest, irritation, pleasure, pride, relief, shame, surprise), but raters were free to add any video likely to induce an intense emotion. This yielded a total sample of 221 clips (.MP4, .MOV, and .WEBM formats), drawn from several media: commercial movies, series, social media, TV shows, and animated films. The majority of excerpts depicted real humans and human interactions.

These six raters then rated each clip on: the label(s) of the emotion(s) felt while watching the video, in a free-labeling task (raters could assign several labels per clip, without restriction), and the valence and arousal, on a 5-point Likert scale using the Self-Assessment Manikin (SAM; Bradley & Lang, 1994). Excerpts were excluded if they were in portrait format, of poor video quality, or longer than 4 minutes. We then applied the criteria of Bachmann et al. (2020) to determine each clip's prevailing label (most-cited label received more than 50% of ratings; second most-cited label, less than 25%). To refine and strengthen this selection, we subsequently enlarged the pool of raters.

Selection

A total of 20 participants (13 women, 7 men), all French native speakers, were recruited from the student population (see Table 2 for demographic information). All provided informed, voluntary consent. Assessment took place online via Teams; participants could not see other raters' responses. Participants reported their current emotional state and any neurological or psychological background. They rated the same features as in the pre-selection phase, with intensity added (10-point Likert scale) and, as a control question, whether they had seen the clip before.

Ratings from this phase were combined with those of the pre-selection phase (which had not assessed intensity). Because label data were categorical, an initial sorting pass was performed: given the heterogeneity of responses, semantically similar labels were grouped, and spelling errors were corrected. Intensity, arousal, and valence ratings were checked for consistency with each excerpt's content. Because these variables are ordinal, the median and median absolute deviation were computed for each excerpt. Results were then derived using the following method:

- First label: occurrences were summed for each label and sorted into semantic clusters (e.g., “revulsion” with “disgust”) to form emotion groups; total occurrences were then calculated per group, and the group with the highest frequency was ranked first and named after its most-cited label. This process was repeated until all groupings were classified.
- Second label: the same method was applied to identify a second, less frequently reported emotion, drawn from a different semantic group than the first label. In practice, the second label rarely differed from the first, with one exception: for the Relief category, participants typically reported anxiety first and relief second, consistent with the temporal dynamics specific to this emotion.
- Ordinal variables: the median and median absolute deviation were computed for valence, arousal, and intensity.
- Prior exposure: the frequency of “yes” responses (having seen the clip before) was computed.

See Table 3 for the results.

Results

Pre-selection

The pre-selection phase yielded a selection of 82 video clips:

Table 1. Results of the pre-selection

Valence	Emotion	Number of videos
Positive	Admiration	5
	Amusement	19
	Happiness	4
	Tenderness	8
Total (positive)		**36**
Intermediate	Relief	3
	Interest	9
	Surprise	1
Total (intermediate)		**13**
Negative	Anxiety	8
	Anger	3
	Disgust	5
	Frustration	1
	Irritation	3
	Discomfort	3
	Contempt	2
	Fear	2
	Sadness	6
Total (negative)		**33**

Category sizes were uneven, ranging from 1 to 21 clips per emotion.

Table 2. Demographic information (selection-phase raters, N = 20)

Characteristic	Value
Gender — Female (%)	65
Gender — Male (%)	35
Age — Mean (years)	21.7
Age — SD (years)	1.61
Education — High school diploma (%)	40
Education — Bachelor's degree (%)	50
Education — Master's degree (%)	10
Emotional state — Happy (%)	40
Emotional state — Stressed (%)	30
Emotional state — Neutral (%)	15
Emotional state — Relaxed (%)	10
Emotional state — Sad (%)	5

Table 3. Results of the selection phase

First label (mean occurrence)	N videos	Second label (mean occurrence)	Intensity (Mdn)	Valence (Mdn)	Arousal (Mdn)	Seen before (mean %)
Admiration (10.33)	3	Surprise (8), happiness (7), amusement (4)	7	4	3	22
Amusement (14.25)	16	Happiness (8.19), disgust (5)	6	4	2	53
Anger (13.37)	8	Discomfort (9), contempt (5), amusement (4.5), happiness (4.5), sadness (4), incomprehension (2)	6	2	2.5	4
Disgust (20.17)	6	Anger (7), amusement (6.5), fear (6), incomprehension (3.33)	6.5	1	3	21
Pride (9)	1	Sadness (8)	6.5	3	3	15
Shame (14)	1	Sadness (4), surprise (4)	5	2	2	0
Interest (14.4)	9	Disgust (7), happiness (4.6), boredom (4.33), admiration (4)	5.5	3	2	13
Happiness (17.64)	14	Amusement (11.33), tenderness (10), admiration (6), relief (5), nostalgia (2)	7.5	4.25	2.5	30
Discomfort (13)	1	Anger (6)	6.5	3	2	0
Fear (19.7)	13	Happiness (11), relief (8.5), disgust (6), surprise (4.63), sadness (4), incomprehension (4), tension (3), anger (3)	5.5	2	3.5	30
Surprise (15)	1	Admiration (4), amusement (4)	5.5	4	2.5	5
Tenderness (5)	1	Sadness (4)	5	3	2	5
Sadness (24.67)	6	Anger (7), empathy (3), disappointment (2)	8	1	3	48
Amusement / joy (11)	1	Surprise (6)	6	4	2	20
Pride / happiness (8)	1	Sadness (3)	6	4	2	21

As a result, a sample of 43 videos likely to induce nine kinds of emotions (amusement, anger, anxiety, disgust, fear, happiness, interest, relief, sadness) was obtained. To balance the distribution of clips across categories, two excerpts from the Schaefer et al. (2010) database were added to supplement the disgust category. To meet the criteria of Rottenberg et al. (2007) regarding the balance of positive- and negative-valence categories, three additional excerpts from the same database were selected to introduce a tenderness category, chosen in accordance with the results reported by Schaefer et al. (2010). Additional information about the selected clips (length, content description) is presented in the Supplementary Material.

The resulting 48 video clips, distributed across ten emotion categories, served as stimuli in the current study.